OSPF Mock Interview – Arrcus Prep

Sriram Dendukuri · 7 October 2026 · 10-question practice round with feedback and model answers

Habit to fix: your answers are correct but too short. For every question use: symptom on each side → why → how I verify (commands) → fix + trade-off.

Scorecard

Q	Topic	Result
1	Neighbour states	Order right; ExStart is master/slave, not DR election
2	MTU mismatch	Right, but too short
3	Inter-area routing	Good; say Type 3 and "distance-vector"
4	Two ABRs / E2 vs inter-area	ECMP right; **E2 vs inter-area wrong** – path type before cost
5	NSSA translation	Flow right; add P bit, highest RID, forwarding address
6	LSA in LSDB, not in RIB	**Missed** – FA reachability, Type 4, AD, filter
7	Options field	New topic; E-bit mismatch = Down, not Init
8	SPF churn	Too short; SPF log, LSA age/seq, throttling
9	DR/BDR	Good; DR reduces adjacencies/flooding, not hellos
10	Area not attached to area 0	Good; add stub-transit restriction and redesign

Revise first: ExStart = master/slave · intra > inter > E1 > E2 · the "LSDB yes, RIB no" checklist.

Q1 – Neighbour states

Question: Two routers on Ethernet in area 0 never reach FULL. Walk through the states and where it could get stuck.

Your answer: Init when configured; 2-Way when own RID seen; ExStart = DR/BDR election; Exchange = DBDs, everyone knows LSDB; Loading = LSRs; Full.

Corrections:

- Down is the first state – no hello received yet.
- Init = I received your hello but my RID is not in it (one-way), not "when configured".
- **ExStart is master/slave negotiation** (higher RID = master, sets DBD sequence number). DR/BDR election happens in **2-Way**, after the wait timer.
- Exchange: DBDs carry **LSA headers only**, not the full LSDB.
- Loading: LSR → LSU → LSAck.

Stuck states – corrected reference notes

Stuck in	What is happening	Causes
Down	No valid hello received; no neighbour entry	Interface down / OSPF not enabled / passive; ACL or CoPP blocking IP protocol 89 or 224.0.0.5; **hello parameter mismatch** (area, hello/dead, mask, auth, E/N bits) – receiver discards the hello
Init	Their hellos arrive but my RID is not listed (one-way)	Unidirectional link; multicast or protocol 89 filtered in one direction; auth on one side only
2-Way	Bidirectional, no adjacency	**Normal between DROthers**; network-type mismatch;

Stuck in	What is happening	Causes
		priority 0 on both
ExStart	Master/slave and initial DBD sequence number	Unicast OSPF blocked (IP 89 – OSPF is not TCP/UDP); duplicate RID; **MTU mismatch** (DBD rejected if its MTU field > own MTU); tunnel/802.1Q overhead
Exchange	DBDs with LSA headers	MTU hidden by mtu-ignore or smaller-MTU device in path drops large DBDs; loss/corruption; bad sequence number or I/MS bits (SeqNumberMismatch → back to ExStart); CPU/memory exhaustion
Loading	LSR → LSU → LSAck not completing	Large LSUs dropped (MTU/fragmentation); unicast reachability or unstable L2; corrupt LSAs (bad checksum)

Why MTU is checked in DBD, not Hello: hellos are small and cannot reveal an MTU problem; DBD starts the reliable exchange, so checking there prevents a FULL adjacency that later black-holes large LSUs. IS-IS instead pads hellos to the MTU.

Q2 – MTU mismatch

Question: A has MTU 1500, B has 9000. What do you see on each router and how do you fix it?

Your answer: A gets stuck in ExStart/Exchange; configure the same MTU.

Missing: why A (rejects DBD whose MTU field 9000 > own 1500); what B shows (retransmits, flaps); how to verify; mtu-ignore trade-off.

> **Model answer:** *A has the smaller MTU, so it rejects B's DBDs because their MTU field is larger than its own, and A stays stuck in ExStart/Exchange. B accepts A's DBDs but keeps retransmitting and never reaches FULL. I'd confirm with show ip ospf neighbor plus the MTU-mismatch log or debug and compare interface MTUs. The fix is matching MTU on both ends; mtu-ignore is only a workaround because large LSUs could then be black-holed.*

Q3 – Inter-area routing

Question: A router in area 1 learns 10.10.10.0/24 that sits in area 2. How is it learned and costed? Is SPF used?

Your answer: Areas must attach to area 0; ABR-2 sends summary into area 0, ABR-1 into area 1; SPF within the area.

Add:

- Name it: **Type 3** summary LSA.
- In area 2 the prefix is in Type 1/2 LSAs; ABR-2 runs SPF to get its cost.
- ABRs **regenerate** Type 3s with their own RID and cost – they do not forward the original.
- Cost = SPF cost to ABR-1 + cost in the Type 3 (e.g. 5 + 30 = 35).
- Between areas OSPF is **distance-vector**; ABRs only use Type 3s received on the backbone – loop prevention.

> **Model answer:** *Inside area 2 the prefix comes from Type 1/2 LSAs. ABR-2 runs SPF and originates a Type 3 into area 0 with that cost. ABR-1 adds its area-0 cost to ABR-2 and originates a fresh Type 3 into area 1. A router in area 1 computes its cost as its SPF cost to ABR-1 plus the Type 3 cost. So SPF runs only within each area; between areas OSPF is distance-vector, which is why every area must attach to area 0 and ABRs only trust summaries from the backbone.*

Q4 – Two ABRs and E2 vs inter-area

Question: Two ABRs advertise the same Type 3 into area 1 – how is one chosen? If the same prefix is also an E2 with a lower metric, which wins?

Your answer: ECMP if equal, else lower cost. External wins because lower metric. **– second part wrong.**

Rule: OSPF compares path type first, then cost: **intra-area > inter-area > E1 (N1) > E2 (N2)**. The inter-area route wins whatever the metric; E2 metrics exclude internal cost and are not comparable.

Different prefix lengths: if the external is /25 and inter-area /24, both are installed and longest-prefix match decides per packet (10.10.10.5 → /25 external; 10.10.10.200 → /24 inter-area). Nothing "wins" – they are different routes.

> **Model answer:** *With two ABRs the router adds its SPF cost to each ABR to the cost in that ABR's Type 3; lowest total wins, equal totals give ECMP. For the external, OSPF prefers route type before metric – intra, inter, E1, E2 – so the inter-area route wins even though the E2 metric is lower.*

E1 vs E2

	E2 (default)	E1
Cost	External metric only – same everywhere	External metric + internal cost to ASBR
Default seed	20 (1 for BGP on Cisco)	Same seed + internal cost
Tie-break	Forwarding metric (cost to ASBR)	Not needed
Use when	Single exit, or exit chosen by external metric	Multiple ASBRs – nearest exit

Example: two ASBRs, metric 20; R is 10 from ASBR-1, 50 from ASBR-2. E2: both 20, tie-break on 10 vs 50 → ASBR-1, route shows 20. E1: 30 vs 70 → ASBR-1, route shows 30. E1 always beats E2 for the same prefix. N1/N2 are the NSSA equivalents. Set with redistribute … metric-type 1.

Q5 – Type 5 vs Type 7 / NSSA

Question: Trace a route redistributed on an ASBR inside an NSSA until a router in area 0 installs it.

Your answer: ASBR converts to Type 7 into NSSA; ABR converts Type 7 to Type 5 into area 0.

- ASBR **originates** a Type 7 – only the ABR translates.
- NSSA blocks Type 5, so Type 7 is the external LSA allowed only inside the NSSA.
- Translated only if the **P bit** is set.
- With several NSSA ABRs, only the **highest router-ID** translates.
- Type 7 always has a non-zero **forwarding address**; the Type 5 keeps it – area 0 routers must reach it.
- No Type 4 needed – the ABR is the advertising router.
- Shows N1/N2 inside the NSSA, E1/E2 elsewhere.

> **Model answer:** *An NSSA blocks Type 5, so the ASBR inside it originates a Type 7 with the P bit set and a forwarding address pointing at itself; NSSA routers install it as N1/N2. The NSSA ABR with the highest router-ID translates it to Type 5, keeping the forwarding address, and floods it into area 0. No Type 4 is needed because the ABR is the advertising router. An area 0 router installs it as E1/E2 if it can reach the forwarding address.*

Q6 – Type 5 in LSDB but not in routing table

Question: show ip ospf database external shows 172.16.0.0/16 but it is not in the routing table. Why?

Your answer: P bit not set by ABR. **– not the cause:** the P bit only governs Type 7 translation; the Type 5 already exists.

#	Reason	How to check
1	**Forwarding address unreachable** – non-zero FA must be reachable via an intra/inter-area OSPF route	FA in show ip ospf database external; show ip route <FA>
2	**No route to ASBR** – FA 0.0.0.0 needs ASBR via Type 1 (same area) or Type 4	show ip ospf database asbr-summary; show ip ospf border-routers

#	Reason	How to check
3	**Better AD** from static/eBGP/connected	show ip route 172.16.0.0
4	**Inbound distribute-list / route-map** filters RIB, not LSDB	show run router ospf; show ip protocols
5	LSA at MaxAge (3600) – being flushed	Age field
6	Router is the ASBR itself	Advertising router = own RID
7	DN bit set (MPLS VPN PE ignores it)	Options field in LSA
8	RIB/prefix limit, corrupt LSA	Logs, show ip ospf statistics

> **Model answer:** *The LSA is in the LSDB, so this is route calculation, not flooding. First the forwarding address: if non-zero it must be reachable via an intra- or inter-area OSPF route. If zero, I need a route to the ASBR via Type 1 or Type 4 – show ip ospf border-routers. Then a better-AD route from another protocol, a distribute-list, MaxAge, or the DN bit on a PE.*

Q7 – Options field

Question: What is the Options field, where is it carried, what do the E and N/P bits do, and what if neighbours disagree on E?

Your answer: Don't know. (In an interview, reason aloud: "it carries capabilities; a stub mismatch blocks adjacency".)

8-bit capability field carried in **Hello** (must match for some bits), **DBD**, and **every LSA header**. Bit order: **DN, O, DC, EA, N/P, MC, E, MT**.

Bit	Meaning
DN	Down bit – MPLS VPN loop prevention; PE ignores LSAs with it set
O	Opaque LSAs (Type 9/10/11 – TE, SR)
DC	Demand circuits
EA	External attributes (obsolete)
N/P	Hello: N = area is NSSA (must match). Type 7: P = translate to Type 5
MC	Multicast OSPF (obsolete)
E	Area carries externals: 1 in normal areas, 0 in stub/NSSA (must match)
MT	Multi-topology

- **E-bit mismatch** (one side stub): both sides discard each other's hellos → no neighbour → **Down** (not Init). Log: mismatched Stub/Transit area option bit.
- Any hello-parameter mismatch → Down. Init = one-direction delivery problem.
- **P bit:** set by the **NSSA ASBR** in the Type 7 LSA header; read by the **NSSA ABR** (highest RID), which translates only if P = 1.
- Don't confuse with the Router LSA V/E/B flags: E = "I am ASBR", B = "I am ABR".

Q8 – SPF running 400 times an hour

Question: Intermittent loss; SPF ran 400 times in an hour. What does it mean, how do you find the source, how do you protect the network?

Your answer: Link flap. (Right direction, too short.)

Cause	Clue
Physical link flap	Interface up/down logs, carrier transitions
Adjacency flap, link up	Dead timer expiring – CPU, congestion, aggressive BFD, unidirectional link
Duplicate router-ID	LSA war – same LSA re-originated with rising sequence; duplicate RID log
Flapping redistributed routes	Type 5 churn (partial SPF, RIB churn)
Adjacency cycling	ExStart ↔ Full repeatedly

Find it: show ip ospf statistics / spf-log (triggering LSA + advertising router) → LSDB entry with very low age and fast-rising sequence → on that router check logs, show interface (carrier transitions, CRC), neighbour uptime.

Protect: SPF throttle (e.g. timers throttle spf 50 200 5000), LSA generation/arrival throttle, IP event dampening, carrier-delay/hold-time, cost out or shut the bad link, iSPF/partial SPF.

> **Model answer:** *400 SPF runs means constant topology change. I'd use the SPF log to see the triggering LSA and advertising router, then the LSDB for an LSA with low age and climbing sequence number. On that router I'd check logs and interface counters for a flapping link, flapping adjacency or duplicate router-ID. Meanwhile SPF/LSA throttling and interface dampening limit impact, and I'd cost out the bad link if there's redundancy.*

Q9 – DR/BDR

Question: R1 is DR; R5 joins with priority 255. Does R5 become DR? What if the DR fails? Why a DR?

Your answer: No; BDR becomes DR and a new BDR is elected; DR reduces hellos. **– last part wrong.**

- Election is **non-preemptive**; R5 becomes BDR when the DR fails, DR only if DR and BDR both fail or the process is cleared.
- **Hellos are not reduced** – everyone still sends hellos to 224.0.0.5.
- DR reduces **full adjacencies** (n(n−1)/2 without DR) and **flooding**: DROthers send LSUs to 224.0.0.6, DR refloods to 224.0.0.5.
- DR originates the **Type 2** network LSA – segment is one pseudo-node in SPF.
- BDR is already FULL with everyone, so failover needs no resync. Priority 0 = never DR/BDR; ties → highest RID.

> **Model answer:** *No, DR election is non-preemptive. If the DR fails, the already-synced BDR takes over and a new BDR is elected – R5. The DR reduces adjacencies and flooding, not hellos: DROthers go FULL only with DR/BDR and send updates to 224.0.0.6, and the DR refloods to 224.0.0.5. It also originates the Type 2 LSA so the segment is one node in SPF.*

Q10 – Area not attached to area 0

Question: Area 3 connects only to area 1. Why can't it reach anything outside? Two fixes, and the restriction?

Your answer: All areas must connect to area 0; virtual link across area 1.

- **Precise reason:** the area 1/3 router has no area 0 interface; ABRs only exchange summaries via the backbone, so Type 3s pass neither way.
- **Fix 1 – virtual link:** area 1 virtual-link <remote-RID> on the area 1/3 ABR and area 0/1 ABR; extends area 0 to the remote ABR; unicast hellos; cost = intra-area cost through area 1.
- **Restriction:** transit area cannot be stub, totally stubby or NSSA (nor area 0).
- **Fix 2 – redesign (preferred):** connect area 3 to area 0, merge into area 1, or GRE tunnel in area 0. Virtual links are a temporary/migration fix.

> **Model answer:** *Area 3's ABR has no area 0 interface, and ABRs only exchange summaries via the backbone, so no inter-area routes pass. The quick fix is a virtual link through area 1 between the two ABRs' router-IDs; area 1 must not be stub or NSSA. The proper fix is redesign – connect area 3 to area 0 or merge it into area 1.*